14th Batch NEU 241 Midterm (Spring 2022)

Credits: Exam Committee Members & 14th Batch

Disclaimer: We cannot guarantee correct answers despite our sincere efforts, please refer back to the lectures to be sure of them. We apologize for any mistakes.

Do not share the document with the doctors!

Exam Committee Heads:
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Q1. A mother had elevated alpha-fetoproteins and polyhydramnios. Which of the following neural tube defects didn't the fetus have?

A. anencephaly
B. rachischisis
C. chiari malformation
D. Encephalocele

Answer: A

Q2. Where can we find Bipolar neurons?

A. Motor neurons
B. Retina
C. Dorsal root ganglion

Answer: B

Q3. A patient needs endotracheal intubation for a long time. What drug should be avoided?

A. Propofol
B. Ketamine
C. Diazepam
D. Etomidate

Answer: D

Q5. A child was born with a cyst at the lumbar region, the cyst was filled with csf and lined with meninges. They had no neurological deficits. Which of the following defects did the child have

A. meningocele
B. meningomyelocele
C. spina bifida occulta
D. Encephalocele

Answer: A

Q6. A question about weakness in the body. With an SMN1 mutation. What is affected?

A. Anterior gray horn
B. Posterior gray horn
C. Spinothalamic

Answer: A

Q7. What tract is affected in Fredreich's ataxia?

A. Spinothalamic
B. Corticospinal
C. Spinocerebellar

Answer: B

Q8. Which staining method is used to visualize the nucleus, cell body, granular endoplasmic reticulum, ribosomes, and number of neurons in a nucleus in the CNS but can not stain the axons?

A. Golgi
B. Weigert
C. Nissl
D. H&E

Answer: C

Q9. What is the mechanism behind the crossed extensor reflex?

A. Polysynaptic - nociceptors
B. Polysynaptic - muscle spindle
C. Disynaptic - muscle spindle
D. Dysynaptic - nociceptors

Answer: A

Q10. Medial branches of the vertebral artery were affected, how will the patient present?

A. Ataxia
B. Vertigo
C. Hemiplegia
D. Dysarthria

E. dysphagia

Answer: C

Q11. A fracture at L4. What structures could be damaged?

A. Cauda equina
B. Femoral nerve

Answer: A

Q12. Tumor at L4. What will the patient present with?

A. Saddle anesthesia
B. Loss of pain sensation at lower limb

Answer : A

Q13. A question asking about the glutamic acid decarboxylase which of the following neurotransmitters does it produce :

A. GABA
B. Glutamate
C. Acetylcholine
D. Dopamine

Answer: A

Q14. If T2 vertebra is affected, Which spinal cord level is affected?

A. T4
B. T6
C. C4
D. T5
E. L1

Answer : A

Q15. CSF goes from lateral ventricle to third ventricle through?

A. Aqueduct of sylvius
B. Interventricular foramen
C. Foramen of luscka
D. Foramen of mengidie

Answer: B

Q16. Morphine is contraindicated in which of the following?

A. intestinal obstruction
B. myocardial infarction
C. Pregnancy
D. Hypertension

Answer: A

Q17. What happens during golgi tendon organ reflex?

A. Alpha neurons inhibit antagonist muscle and stimulate agonist muscle
B. Alpha neurons inhibit agonist muscle and stimulate antagonist muscle
C. Gamma neurons inhibit antagonist muscle and stimulate agonist muscle
D. Gamma neurons inhibit agonist muscle and stimulate antagonist muscle

Answer: B

Q18. What can be given before conception to prevent neural tube defects?

A. Folate
B. Valproic acid
C. Increased vitamin A
D. Iron

Answer: A

Q19. Damage to the cerebral peduncle in the midbrain damaged the corticobulbar tract. What will be the presentation?

A. Paralysis of the full contralateral face
B. Paralysis of the lower two thirds of the contralateral face
C. Contralateral hemiplegia
D. Contralateral weakness

Answer: B

Q21. A case about a patient that presented with a hemorrhage that appeared convex in shape on MRI, the artery causing this hemorrhage exit through what foramen?

A. Foramen lacerum
B. Foramen magnum
C. Foramen spinosum
D. Foramen magnum

Answer: C

Q22. What causes sensitization of nociceptors

A. substance P
B. Prostaglandins
C. histamine

Answer: B

Q23. The formation of cerebrospinal fluid by the choroid plexus includes (1) osmosis of water, (2) active transport of sodium, and (3) passive diffusion of chloride. Which sequence best describes the, correct temporal order of these processes?

A. 2, 3, 1
B. 3, 2, 1
C. 1, 3, 2
D. 3, 1, 2
E. 1, 2, 3
F. 2, 1, 3

Answer: A

Q24. Which of the following is a potent Analgesic and anesthetic?

A. Propofol
B. Ketamine
C. Fentanyl
D. Meperidine

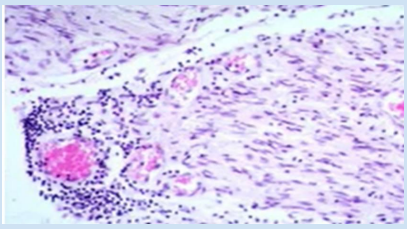
Answer: A

Q25. What is a migraine treatment that works on 5HT-1B/D?

A. Sumatriptan
B. Dihydroergotamine
C. Lasmiditan
D. Gepants

Answer: A

Q26. there was one about ascending paralysis and on histo slide you see perivenular lymphocytosis?



A. Segmental demyelination
B. Axonal dystrophy
C. Distal axonopathy
D. Wallerian Degeneration

Answer: A

Q27. Pregnant woman needing migraine treatment, which one is contraindicated?

A. Sumatriptan
B. Rizatrpitan
C. Dihydroergotamine
D. Gepant

Answer: C

Q28. What receptor receives high vibrations?

A. Merkel cells
B. Meissner's corpuscle

C. Pacinian corpuscle
D. Ruffini endings

Answer: C

Q29. Which one is the metabotropic receptor of the following?

A. Glycine
B. P2X
C. D1

Answer: C

Q30. A 5 year old child presents with a mass in the 4th ventricle and shows enlarged lateral and 3rd ventricles. The microscopic examination reveal's peri vascular rosettes, what is the diagnosis?

A. Astrocytoma
B. Medulloblastoma
C. Ependymoma
D. Glioblastoma

Answer: C

Q31. A girl has trouble falling asleep. What drug should be used?

A. Triazolam
B. Midazolam
C. Clonazepam

Answer: A

Q32. Question asking about what is the biggest driving force for resting membrane =-95?

ECl=-100
EK=-70
ENa=+30
ECa=+95

A. Cl

B. K
C. Na
D. Ca

Answer: D

Q33. A patient comes with Tabes dorsalis and syphilis. Dorsal roots and are affected and he has areflexia. What would be affected?

A. Vibration on the ipsilateral side
B. Pain on the ipsilateral side
C. Temperature sensation on the ipsiateral side
D. Vibration on the contralateral side

Answer: A

Q34. Question had an image where fasciculus gracilis was labeled. Which of the following presentations are correct?

A. Ipsilateral Vibration of the leg
B. Ipsilateral Position sense of the leg

Answer: Both A + B were accepted

Q35. A person woke up with homonymous hemianopia. Further investigations show occlusion in artery which supply the posterior limb of the internal capsule, thalamus, hippocampus, and optic chiasm. Which artery is occluded?

A. Medial lenticulostriate
B. Anterior choroidal
C. Posterior cerebral artery
D. Thalamostriate artery

Answer: B

Q36. A Physiology question about muscle weakness and myositis and no antibodies against acetylcholine receptor and tyrosine kinase and there is a lung tumor, which of the following is impaired?

A. Ca+2 receptors on the presynaptic

B. Na+ receptors on the postsynaptic
C. K+ receptors on the postsynaptic
D. SNARE

Answer: A

Q37. Right homonymous hemianopia. Infarction in blood supply to the hippocampus, tail of caudate, and amygdala. What artery is affected?

A. Anterior choroidal artery
B. Posterior cerebral artery
C. Middle cerebral artery
D. Basilar artery

Answer: A

Q38. A 63-year-old man has been receiving hemodialysis for chronic renal failure and has noted increasing loss of sensation in his legs for the past 4 years. On physical examination, there is symmetrically decreased sensation over both lower extremities. He has no decrease in strength or abnormality of gait. What is most likely to produce these findings?

A. Guillain-Barré syndrome
B. Cerebral astrocytoma
C. Cerebral infarction
D. Diabetes mellitus
E. Multiple sclerosis

Answer: D

Q39. A 46 year old man presents with focal calcification in the cerebral hemisphere and high levels of GFAP. What is the genetic mutation?

A. BRAF
B. 1p 19q co deletion
C. C-myc
D. NF2

Answer: B

Q40. A doctor is testing dorsiflexion of the foot. What nerve is being tested?

A. Deep fibular nerve
B. Superficial Fibular nerve
C. Tibial nerve
D. Femoral nerve

Answer: A

Q41. What drug should be used to prevent malignant hyperthermia?

A. Sevoflurane
B. Isoflurane
C. Desflurane
D. Propofol
E. Etomidate

Answer: A

Q42. Which of the following signs of nerve injuries caused by the fracture shown in the following X-RAY:

A. Hand of benediction
B. Claw hand
C. Erb's-palsy
D. Klumpke's palsy

Answer: B

Q43. What is the mechanism of vincristine?

Answer: Affecting the cytoskeleton of axons

Q44. Glutamate binds to mGLUT and works through IP3 and DAG. What type of G-protein?

A. Gs
B. Gq
C. Gi

Answer: B

Q45. A patient comes with a right dry eye. What is the affected nucleus?

A. Inferior salivatory nucleus
B. Superior salivatory nucleus
C. Edenger-Westphal
D. Abducens nucleus

Answer: B

Q46. A stroke to the posterior part of the parietal lobe. What symptoms will the patient have?

A. Alexia with agraphia
B. Apraxia

Answer: A

Q47. The basilar artery is occluded. What will the patient have?

A. Medial strabismus
B. Quadriplegia
C. Unilateral anesthesia

Answer: B

Q48. What is the reason for the loss of pain and temperature in central cord syndrome. Damage to:

A. crossing fibers of the spinothalamic tract
B. dorsal column
C. anterior gray horn
D. posterior gray horn

Answer: A

Q49. A newborn is a floppy baby. With an SMN1 mutation. What is affected?

A. Anterior gray horn

B. Posterior gray horn
C. Spinothalamic

Answer: A

Q50. The parotid gland is supplied by a parasympathetic nerve from CN 9 which passes in which of the following ganglions?

A. Otic ganglion
B. Inferior salivatory
C. Superior salivatory

Answer: A

Q51. What receptor does lasmiditan act on?

A. 5HT-F
B. 5HT-B

Answer: A

Q52. What is the mechanism of action of antidote benzodiazepines?

A. Indirect agonist of GABA
B. Competitive antagonist of GABA

Answer: B

Q53. Patient developed right sided ptosis and lateral strabismus after a stroke. He also had left sided spastic paralysis. MRI would show a stroke involving which part of the brain?

A. Left pons
B. Right medulla
C. Right midbrain
D. Left midbrain
E. Right pons

Answer: C

Q54. A boy was cooking. He touched a hot element on the oven and immediately withdrew his hand, but he burned his fingertip. Later he put water (30° C) on the burned area which was swollen and developed erythema and heat. What causes the area around the site of injury to be more sensitive to pain?

A. Acetylcholine
B. Substance P
C. Serotonin
D. Glutamate

Answer: B

Q55. Question about Ach in NMJ and what its properties?

Answer: the receptor allows both Na+ and K+ movement however due to its Erev it causes Na+ to influx only and depolarize.

Q56. A patient is suffering from an aortic aneurysm. What will be the most likely presentation?

A. Central cord syndrome
B. Anterior Cord syndrome
C. Tabes Dorsalis
D. ALS

Answer: B

Q57. A picture of a doctor testing for the dorsiflexion of the foot was shown. Which of the following nerves is being tested?

A. Tibial
B. Deep fibular
C. Femoral

Answer: C

Good Luck and may Allah reward your tremendous hard work and grant you success in all aspects of life!

Please keep us in your prayers <3